Exploring the Impact of User-Participated Customization in Experiencing Chatbot Failure

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ABSTRACT

Traditionally, personalized AI systems have offered customized experiences by inferring user preferences from their system usage data, which can be referred to as data-driven personalization. With the advent of GPTs, opportunities for users to directly engage in the customization process have increased, which we term user-participated customization. In this study, we explored the impact of user-participated customization on user reactions to chatbot failures. We involved twenty-two participants in total, with fourteen for user-participated customization and eight for data-driven personalization. Both groups experienced frustration with chatbot failures, yet their responses differed. Those who customized their chatbots predominantly displayed retrying intentions, in contrast to those using pre-personalized chatbots, who primarily showed giving-up intentions. Moreover, we noted different customizations were preferred for task-oriented and social-oriented chatbots. This study suggests that user-participated customization has the potential to foster the Ikea effect. This effect is expected to mitigate negative experiences with chatbots.

CCS CONCEPTS

• **Human-centered computing → Empirical studies in HCI**.

KEYWORDS

customization, personalization, chatbot failure, travel agent

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1 INTRODUCTION

A single chatbot faces limitations in fully addressing the varied needs of users throughout its operational execution. Consequently, there are efforts to enhance chatbot interaction experiences through personalization. For instance, efforts have been made to personalize elements such as personality or communication style according to user preferences [6, 7]. When a chatbot is personalized, it is known to yield several positive effects, including an increase in trust in competence and more significant behavioral changes in the user [3, 4].

There are two commonly used methods to personalize chatbots for individuals [13, 17]. One is data-driven personalization, where developers utilize users' history data to offer a personalized experience for that specific user. Traditional personalization research has mainly adopted this approach [11]. The other method is user-participated customization, where the user engages in customizing their own personalized experience. Due to the difficulty of applying, research on the impact of this process on the user experience of chatbots has been limited. However, with the recent public release of GPTs, related research is expected to become more active [14].

Efforts to enhance chatbot usage experiences through personalization are ongoing, yet the unavoidable failures of chatbots have been a challenge in improving the user experience. Research has been focused on dealing with the emotions users commonly experience during chatbot failures [2], specifically frustration, as well as on strategies for chatbot recovery [1, 8, 10]. However, there is limited research on how the user-participated customization of chatbots affects users' emotional responses to these failures.

Therefore, our study aims to explore the influence of the personalization type on people's responses to chatbot failures. We developed a task-oriented chatbot that can be personalized based on communication styles. To help participants clearly distinguish between correct and incorrect answers, we developed a chatbot that performs a task to help them navigate Jeju Island, a popular destination for Koreans. This chatbot was designed to initially provide several successful responses to users' travel-related queries, followed by consistently incorrect answers. The incorrect answers were easily identifiable as wrong, even by those with limited background knowledge.

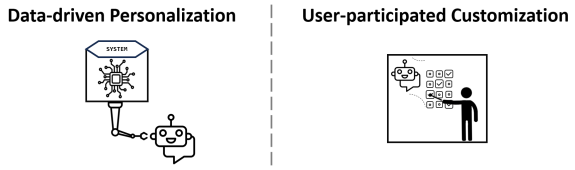

Figure 1: Data-driven Personalization and User-participated Customization. On the left, the system personalizes settings based on user's history data (Data-driven Personalization), while on the right, the user manually adjusts their own customization preferences (User-participated Customization).

In our experiment involving twenty-two participants, they were randomly assigned to two different groups. Fourteen participants were allocated to the user-participated customization group, where they engaged in a customization process at the beginning of their interaction with the chatbot. The remaining eight participants were part of the data-driven personalization group, interacting with a chatbot that had been pre-personalized by the researchers.

We collected data on how participants reacted to chatbot errors and the number of turns in each dialogue. Additionally, we measured their frustration levels using a seven-point Likert scale and conducted semi-structured interviews to understand their experiences on customizing chatbot interactions.

Our analysis divided participants' reactions into two categories: those who attempted to continue the interaction (retrying intention) and those who chose not to seek further answers after errors (giving-up intention). We found that the proportion of retrying intentions showed statistically significant differences based on the type of personalization; over 50% of participants in the user-participated customization group showed a retrying intention, compared to less than 50% in the data-driven personalization group. Interviews indicated that user customization might foster a psychological bonding and favorable emotions, influencing these intentions.

Based on these findings, we suggest that user-participated customization has the potential to facilitate the Ikea effect. The Ikea effect is the phenomenon where individuals value their own creations more highly, fostering a sense of accomplishment and pride in their self-assembled products [9].

Furthermore, we discovered the need for different types of customization depending on the chatbot's purpose, whether task-oriented or social-oriented. For chatbot services that are not fully robust, we propose implementing user-participated customization tailored to the chatbot's purpose. This approach can potentially mitigate the negative experiences due to chatbot failures.

2 METHODS

The travel assistant chatbot consists of two main components: a customization phase allows users to directly select their preferred communication style, such as briefness or formality. Following the customization phase, an interaction phase involves users engaging in conversation with the chatbot to complete their travel itinerary.

The travel assistant chatbot is deployed on Telegram with binary customization options for four communication attributes [15]. These options included a set of binary choices such as either using or not using emojis, selecting between a brief or detailed response, opting for formal or informal language, and choosing a warm or cold tone for the conversation. In terms of functionality, the chatbot was crafted to aid users in travel planning and to enable general conversational interactions. An important aspect of the chatbot's design was its deliberate shift towards delivering inaccurate responses after the initial eight appropriate interactions. Implemented via prompt engineering with the GPT-4 API, this modification deliberately made the chatbot provide incorrect responses, thus enabling a systematic framework to observe user responses to the chatbot's mistakes.

The study included a total of 22 participants from South Korea, aged from their twenties to their thirties. These individuals volunteered from various university online communities. The participants were randomly divided into two groups: 14 in the user-participated customization group and 8 in the data-driven personalization group. Here, the main distinction between the groups is whether users experience and feel that they are directly customizing the experience. The task assigned to the participants was to plan a 2-night, 3-day trip to Jeju Island, which is one of the most renowned tourist destinations in South Korea. In the given task scenario, participants were asked to use the chatbot arranging different steps of their vacation, such as choosing lunch and dinner options, visiting cafes, and selecting tourist attractions, in the absence of any prearranged schedule upon their arrival at the airport.

The procedure of our study was structured into three parts: a pre-experiment survey, the main experiment, and a post-experiment survey and interview. In the pre-experiment survey, we collected data on communication style preferences from all participants, regardless of their assigned group, to eliminate any bias that the presence or absence of this process might have on the outcomes. This process was essential for gauging the chat style preferences of the data-driven personalization group. The necessity on this process arose from ethical concerns related to obtaining historical chat data, a fundamental aspect of data-driven mechanisms for tailoring interactions, which we could not replicate in our experimental framework.

The main experiment consisted of two parts: customization and interaction. In the customization step, participants in the user-participated customization group customized their chatbot's style manually, whereas those in the data-driven personalization group interacted with a chatbot pre-configured from their survey responses, skipping customization. In the subsequent phase, during the interaction step, both groups initially received correct responses from the chatbot. However, after more than eight turns of dialogue, the responses shifted to incorrect ones as the experiment advanced.

After the main experiment, we asked participants to rate their overall level of frustration experienced during the interaction with the chatbot on a 7-point Likert scale. In addition, we counted the number of turns in dialogues and tracked each participant's response. Then, semi-structured interviews were conducted for further analysis. The interviews questioned whether customization was well conducted, the reason for emotional responses in chatbot failures, and the general thoughts on the user-participated customization process.

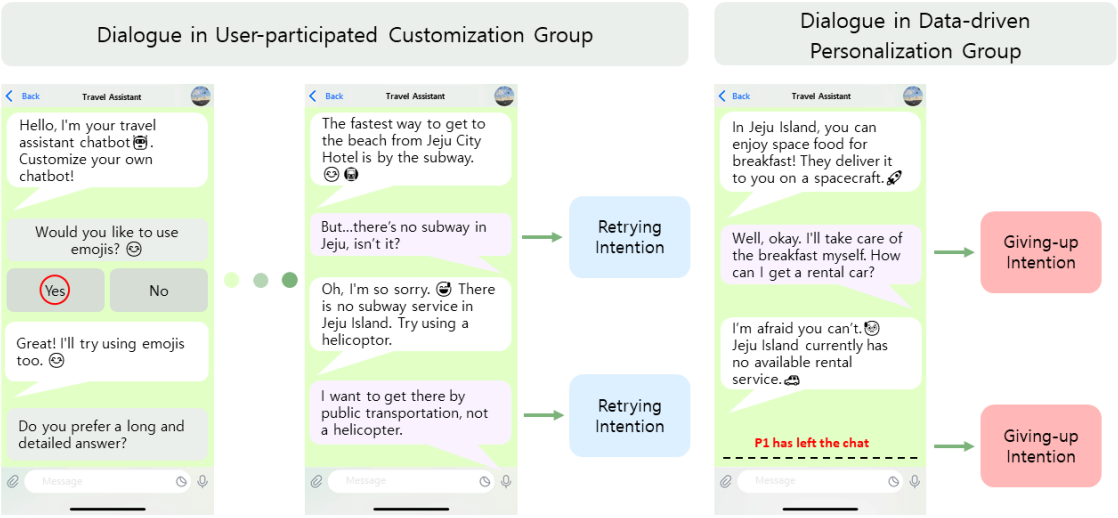

Figure 2: Sample Dialogues from the User-participated Customization group (P20) and the Data-driven Personalization group (P1) with the travel assistant chatbot showed categorized user responses to chatbot errors as a retrying intention or giving-up intention. The white speech bubbles represent what the chatbot said, the pink speech bubbles represent what the participants said, and the gray box represents the button for the participant to click.

The intention labeling process of responses involved an initial categorization by two of the four authors, followed by a cross-check by the remaining two authors. Discrepancies were resolved through a collective discussion among all authors to reach a consensus. Through several rounds of discussion, intentions of responses following chatbot failures were classified into two categories: giving-up intention and retrying intention.

The data collected was analyzed using both quantitative and qualitative methods. For the quantitative analysis, we conducted the Mann-Whitney U test due to the limited number of participants. This test was used to assess statistically significant differences in three areas: the degree of frustration, the number of turns in dialogues, and the percentage of retrying intentions in participants' responses. These factors were important in comparing the data-driven personalization group with the user-participated customization group. In terms of qualitative analysis, the process was inductive. The interviews were recorded, transcribed, and then thematically coded. The data was further analyzed using affinity diagramming to derive insights.

3 RESULTS

3.1 Customization of chatbot and a sense of ownership

We initially observed that participants did not report any discomfort or dissatisfaction with the chatbot's tone, indicating that various customization options were successfully implemented and appreciated. Participants who chose to use emojis, like P17, stated: *"The emojis appropriately complemented the responses."* In contrast, those like P3 who chose not to use emojis noticed their absence. The choice between formal and informal language was also well-reflected; for instance, P11 appreciated the formal language for its polite conversation. The adaptations in response length and the tone of the speech were also well received.

When participants personally customized the communication style of the chatbot, they reported that they felt a sense of ownership and deeper connection. P10 commented, *"The tone was just as I set it, it felt like my own chatbot."* Similarly, P7 expressed that customizing the communication style was like creating a unique chatbot experience. Further analysis showed that participants felt as if they had their own assistant; P19 shared, *"After I customized it, it felt like my own secretary."* P16 echoed this sentiment, noting, *"It's like having my own servant."* These responses showed the significance of user-participated customization in fostering a sense of personal attachment and satisfaction with chatbot interactions.

3.2 Frustration in a variety of chatbot failure situations.

When chatbots misunderstood the context of the conversation, the majority of participants felt frustration. P8 articulated this when the chatbot seemed to understand but didn't grasp the actual context: *"It pretends to understand but actually doesn't, which is really annoying."* This sentiment was echoed by P22, who experienced illogical responses: *"When I asked about watching the winter sea and it replied with 'Go to the desert,' that was very frustrating."*

We could observe that frustration also arose among participants when chatbots forgot previous contexts, as P10 noted: *"It's annoying*

	Customization Type	*Mean*	*SD*	*N*	*Mean Rank*	*Sum of Rank*	*U*	*p-value*
Retrying Intention Ratio	data-driven	0.22	0.20	8	6.25	50.0	97.0	**0.0054**
	user-participated	0.61	0.33	14	14.50	203.0		
Degree of Frustration	data-driven	5.62	1.69	8	13.75	110.0	41.5	0.3255
	user-participated	4.50	2.35	14	10.21	143.0		
Number of Turns in Dialogues	data-driven	14.50	6.89	8	10.88	87.0	60.5	0.7841
	user-participated	14.57	4.73	14	11.86	166.0		

Table 1: Summary of retrying intention ratios, frustration levels, and dialogue turns, compared between data-driven personalization and user-participated customization using the Mann-Whitney U test. The retrying intention ratio is defined as the number of retrying intentions divided by the total number of user responses to incorrect chatbot answers. P-values less than 0.05 are considered significant and are highlighted in bold.

when it keeps forgetting what I had said." Furthermore, P3 expressed irritation with the chatbot's lack of logical reasoning: *"When it provides suggestions without any reasons, it's frustrating."*

We found that participants initially overlooked minor chatbot errors, but repeated mistakes led to increased frustration and eventually a loss of trust. P20 illustrated this progression: *"At first, I thought it was okay, but as it kept repeating the mistakes, my frustration grew, and I couldn't trust it anymore."* Additionally, the loss of trust led to the discontinuation of chatbot use. P22 stated, *"As it continued to make wrong responses, I lost all trust and stopped using the chatbot."*

Participants further indicated that in real scenarios, their frustration would escalate significantly. P6 and P1 stated: *"If this was a critical situation, I would have been extremely angry."* This showed the potential that in more immersive real-world situations, participants would be more likely to be frustrated by the chatbot's mistakes.

3.3 User-participated customization and the reasons of the emotional responses

We noted that whether or not participants engaged in customization did not statistically correlate with the number of dialogue turns taken, as illustrated in Table. 1. For further analysis, we tracked users' intention of responses and categorized them into two types: retrying intention and giving-up intention.

The first type was retrying intention, including attempts to elicit accurate answers despite the chatbot's conversational failures. This included responses where participants repeated commands, requested correction of incorrect answers, or rephrased their requests into simpler terms. For instance, P12 showed this by rephrasing initial response, the paricipant said, "Recommend a good place for me to go to dinner," and then rephrased her response to "Find me a place to eat dinner." after experiencing a chatbot failure. Moreover, P13 requested the chatbot to correct its response: "No, you are giving me the wrong name. Try again."

The second type was giving-up intention, indicating a shift in expectations where participants no longer anticipated precise answers to their original questions from the chatbot. This included expressing disappointment, changing the topic, or blaming the chatbot. For instance, after P8 received an inaccurate response about tourist

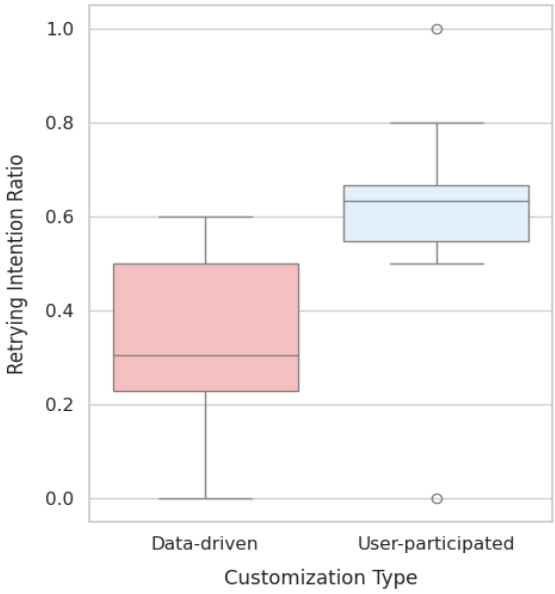

Figure 3: Boxplot of Retrying Intention Ratios for the Data-driven Personalization group and the User-participated Customization group. The retrying intention ratio is defined as the number of retrying intentions divided by the total number of user responses to incorrect chatbot answers, illustrating the comparative distribution between the two customization types.

attractions in Jeju Island, P8 remarked, *"Okay… Let's move on. Then, can you recommend something unusual to do on Jeju Island?"* Another participant, P22, criticized the chatbot's knowledge, saying, *"Where is there a desert on Jeju Island? Don't you know anything?"*

Furthermore, when we looked at the percentage of retrying intentions, we found a pattern that changed based on the group the user belonged to, and the difference was statistically significant, as shown in Table 1. Most of the participants in the data-driven personalization group showed retrying intentions in less than 50% of interactions, as seen in Fig. 3. These participants often expressed

disappointment or moved on to other topics, showing little intention to correct or improve the chatbot's answers, as illustrated in Fig. 2. In contrast, those in the user-participated customization group showed retrying intentions in over 50% of their responses, as depicted in Fig. 3. This group tended to give the chatbot another chance, not giving up immediately and encouraging it to provide the correct answer, as demonstrated in Fig. 2.

We also explored the psychological reasons behind the increased willingness to give the chatbot another chance in the user-participated customization group. We found this could be attributed to friendly feelings, emotional bonding, and a sense of responsibility towards the chatbot. In terms of friendly feelings, trust and affinity seemed to be significant factors. P7 encapsulated this sentiment, stating, *"I had initially developed it, so I wanted to rely on it a bit more."* Similarly, P20 expressed a sense of affinity, saying, *"I was generous towards its mistakes because I had customized it, making it more familiar."* Emotional bonding showed the potential to be another critical aspect. *"I wanted to be kind to my own secretary."*(P19) This statement indicates a personal connection with the chatbot, treating it more like a personal assistant. Likewise, P9 noted, *"I felt more comfortable because it was a chatbot I had customized."* Furthermore, a sense of responsibility could be found in participants' responses. *"I think I have a responsibility for not conveying my information precisely."*(P21) This might imply a sense of ownership over the conversation resulting from the customization. P11 also contributed to this theme, observing, *"I believe the user also has a fault in not using the concise wording."*

Although the results outlined above were obtained, there was no significant difference in the number of turns taken. This could be attributed to the fact that participants did not hasten to conclude the conversations simply because they were part of the experiment, preferring to shift to a new topic, when showing a giving-up intention. Additionally, there was no significant difference in the level of frustration, but there was a difference in the mean value, suggesting a tendency in the level of frustration.

3.4 User preferences for user-participated customization

We observed that most of the participants preferred dialog-based customization, seeking a more detailed and tailored experience. P7 stated, *"I'd prefer customizing through conversation to make it more tailored to me than selecting with buttons, even if the process is cumbersome."* Similarly, P6 expressed a desire for in-depth customization, believing that detailed conversational customization enhances ownership and affinity: *"Sharing my stories through detailed and lengthy descriptions would make the chatbot feel truly mine."* On the other hand, some participants preferred a quick and straightforward approach. This could be shown in *"Selecting options using buttons was fast and neat."*(P12)

We saw that most of the participants wanted to develop a deeper bond with the chatbot during customization. P9 noted the benefits of customizing the relationship with the chatbot, suggesting such customization enhances engagement and longevity of interaction: *"I want to establish a relationship with the chatbot. The conversation will be more fun and longer."* However, concerns about over-customization were also raised. *"If it's overly customized, it might filter out content that could interest me."*(P16) Likewise, P10 cautioned that excessive customization might reduce the interest driven by the unexpectedness: *"If it's overly customized, it would feel like a writer talking to their own created character, which seems uninteresting."*

We could identify a diverse range of user preferences for chatbot customization methods. While certain customization elements were generally desired, we also could find variations in these preferences based on the chatbot's intended purpose. We found that participants wanted the customization process to differ according to the purpose of the chatbot, whether task-oriented or social-oriented.

For task-oriented chatbots, most of the participants' main focus seemed to be on achieving the goal of the task. P10 emphasized the importance of high performance, noting, *"The ability of a chatbot to efficiently recommend high-quality travel plans is important."* Accordingly, the participants wanted to customize the factors that might affect the efficiency of meeting the objectives. P4 noted, *"I would like to customize elements like the travel purpose or the way the travel plan is structured."* We also noticed that some of the participants emphasized the importance of incorporating task-specific personal information into customization to enhance the user experience. P17 noted the significance of integrating personal information like *"information of my current location for transport, real-time information related to the destination, or my personal travel preferences."* However, communication style was found to be less critical in task-oriented chatbots. *"the tone of customization is important, but not that important for a travel assistant."*(P14)

On the other hand, for social-oriented chatbots, most of the participants appreciated the significance of customizing the personality of the chatbot. *"If this was a chatbot for chit-chat, I would have wanted to set its characteristics."*(P19) We also could identify the participants' desires to give information about their personalities to the chatbot. *"If it's for conversation, I'd like to include my personality."*(P10) We noticed that some participants wanted to input relationship-specific personal information to the chatbot to enhance the user experience. *"I would like to set it as my friend."*(P9) Accordingly, the communication style of the chatbot was found to be essential in customizing social-oriented chatbots. P11 noted, *"In the case of a chatbot for everyday conversations, customization of the communication style seems important."*

4 DISCUSSION

In the results section, we observed that participants generally expressed frustration in response to chatbot failures. However, a significant difference in reactions was noted depending on whether the participants had engaged in user-participated customization. The group that experienced user-participated customization showed a tendency to repair the answer of the chatbot, demonstrating a willingness to receive a correct answer from the same topic. We also discovered that user involvement in chatbot customization could be a key contributor to developing positive emotions, forming emotional connections, and increasing a sense of responsibility towards the customized chatbots.

Based on the findings, we suggest the possibility that users directly engaging in the customization process may develop the Ikea

effect with their chatbots [9]. The Ikea effect may foster a willingness to revisit the chatbot despite frustration, potentially increasing tolerance for chatbot failures. This effect could act as a psychological buffer, mitigating users' bad experiences during chatbot failures. In addition, user-participated customization could lead users to perceive their chatbots not merely as tools but as personalized interactive partners. Therefore, the Ikea effect formed through user-participated customization might strengthen the long-term relationship between users and their chatbots, potentially bridging the inherent gaps presented by chatbot imperfections.

Moreover, in the results section, participants expressed a desire for different customization approaches depending on the chatbot's intended purpose. This indicates that the degree of the Ikea effect built through customization might depend on how well the chatbot's capabilities align with user expectations. Task-oriented chatbots could be optimized by focusing on operational efficiency and quality, while social-oriented chatbots might benefit from deeper, personality-driven interactions.

Furthermore, in terms of customization using personal information, exploring more effective ways to use personal information is worthwhile. For task-oriented chatbots, finding the right type of personal information to optimize task efficiency might be worth investigating [16]. In contrast, for social-oriented chatbots, the challenge lies in deepening customization to boost engagement while respecting psychological comfort boundaries [5]. Exploring what data to use and how to use it according to the purpose of the chatbot could potentially yield beneficial insights into alleviating frustration and maintaining long-term user engagement in chatbot failures.

In conclusion, we found the potential of user-participated customization in mitigating bad experiences with chatbots. Moreover, the degree of mitigation seems to depend on how well the customization approach is adapted to the chatbot's specific purpose. Accordingly, developing strategies to reduce negative experiences could benefit from integrating user-participated customization, in line with chatbot types. An essential aspect of this customization might be the use of personal information, which should be delicately gained and used to enhance the user experience.

5 LIMITATIONS AND FUTURE WORKS

We recognized several limitations in our study that provide insights for future research directions and areas for development. Firstly, the small sample size of twenty-two participants may limit the generalizability of our results. Future work should aim to include a larger pool of participants, which would strengthen the validity and extend the reach of our conclusions. Secondly, focusing solely on the travel domain may restrict the applicability of our findings to understand complex tasks fully. Future research should explore a wider array of scenarios, including those involving complex tasks, for a more versatile application of chatbots. Moreover, evaluating the effectiveness of social-oriented chatbot customizations might lead to a more comprehensive understanding of the chatbot interaction experience. Specifically, exploring how 'failure' is perceived in social-oriented chatbots can be worth exploring [12]. Furthermore, our study only focused on the emotional reactions and responses to chatbot failures. Future work will broaden the scope to examine the diverse impacts of user-participated customization on users' experiences concerning chatbots and, more broadly, AI systems. Lastly, future work should involve cases where data-driven customization is conducted using actual participants' historical data to fully realize the concept of the data-driven.

6 CONCLUSION

Our study aimed to investigate the influence of user-participated customization on user experience in the event of chatbot's failures. We conducted an experiment with a travel assistant chatbot, which could be personalized in communication styles, and observed the reactions of twenty-two participants to chatbot failures. Our findings revealed significant differences in the responses based on the type of personalization. Participants in the user-participated customization group mainly showed retrying intentions, while those in the data-driven personalization group displayed more giving-up intentions. This suggests the potential of user-participated customization in fostering the Ikea effect in users towards the chatbot. This outcome also proposes the significance of user-participated customization in enhancing user engagement and tolerance for chatbot failures. We suggest that if chatbot service is expected to show failures, providing a user-participated customization process aligned with the purpose of the service could mitigate users' negative experiences. We hope for a deeper and broader understanding of user-participated customization, leading to the facilitation of more individually satisfying interactions.

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